

Fig. 4: Actual-size, single-side PCB for the remote control

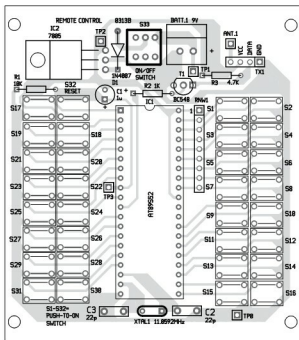


Fig. 5: Component layout for the PCB in Fig. 4

and RL2) and a few discrete components.

Power-on reset is provided by the combination of resistor R4 and capacitor C4. Switch S34 is used for manual reset. Crystal X_{TAL2} along with two 22pF capacitors provides basic clock frequency to microcontroller IC4. Connectors CON1 through CON31 are con-

nected to input/output pins of IC4, except port pin P3.0 (RXD). Port pin P3.0 is used for signal reception.

Microcontroller IC4 decodes the received signal and controls the relays through its port pins. Port pin P0.0 goes low when switch S1 is pressed at the transmitter end. Port pin P0.0 provides clock signal to flip-flop IC5. Flip-flop is used in toggle mode by connecting output $\bar{Q}$ to its data pin D. The logic level at D input is transferred to Q output during positive-going transition of the clock pulse. High output Q1 of IC5 drives transistor T3 into saturation and energises relay RL1. The appliance is connected through contacts of the relay.

To get power supply for the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 1A. The transformer output is rectified by a full-wave rectifier comprising diodes D2 through D5, filtered by capacitor C7 and regulated by IC 7805 (IC3). Capacitor C8 bypasses ripples present in the regulated supply. LED1 acts as the power indicator and R7 limits the current through LED1.

Software

Source programs for microcontrollers of the transmitter and receiver sections are written in 'C' language and compiled using Keil μ Vision4 compiler. The generated hex code is burnt into

PARTS LIST

Semiconductors:

IC1, IC4	- AT89S52 microcontroller
IC2, IC3	- 7805, 5V regulator
IC5	- CD4013 dual flip-flop
T1-T4	- BC548 npn transistor
LED1	- 1N4007 rectifier diode
TX1	- 5mm LED
TX1	- ASK transmitter module
RX1	- ASK receiver module

Resistors (all $\frac{1}{4}$ -watt, ± 5 per cent carbon, otherwise stated):

R1, R4	- 10-kilo-ohm
R2, R5, R8, R9	- 1-kilo-ohm
R3, R6	- 4.7-kilo-ohm
R7	- 330-ohm
RNWL, RNW2	- 10-kilo-ohm resistor network

Capacitors:

C1, C4	- 10 μ F, 16V electrolytic
C2, C3, C5, C6	- 22pF ceramic disk
C7	- 1000 μ F, 25V electrolytic
C8	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 9V, 1A secondary transformer
X_{TAL1} , X_{TAL2}	- 11.0592MHz crystal
S1-S32, S34	- Push-on tactile switch
S33, S35	- Push-on toggle switch
RL1, RL2	- 12V, 1C/O relay
BATT.1	- 9V
CON1-CON31	- 2-pin male connector
CON1(F), CON2(F)	- 2-pin female connector

the microcontroller by using a suitable programmer.

In the transmitter source code, you will see that for every switch pressed there is a particular ASCII code transmitted via TXD pin of the microcontroller. For serial communication, TMOD register is loaded with 20H for 8-bit auto reload mode, which is well suited for most UART transmissions. SCON register is loaded with 50H for 8-bit data frame with start and stop bits. Then TH1 is loaded with F4H for 2400 baud rate at 11.0592MHz crystal.

Now for each switch pressed, an ASCII code is generated. The code is not sent directly to transmitter module. A very common and well-known error-control coding technique is used for TX-RX pair transmission. Function 'sendsignal' is responsible for data transmission.

The frame generated for each data is:

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'9' '8' '8' 'Signal' 'Inverse of Signal' '7'
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where '9' is junk data to stabilise the RX module. '8' is frame start, another '8' is second frame, 'signal' is the original ASCII code, 'inverse of signal' is